Probabilistic digit recognition with evidential uncertainty

A local adaptation of a WACV 2024 computer vision method

Report prepared: 6 October 2026. All code, experiments, figures, and report text were generated by the AI agent. Personal fields are intentionally blank at the student's request.

Student name: _____
Student ID: _____

1. How AI found the algorithm

The agent first extracted and visually inspected AIassign2.pdf, then recorded its six report requirements and instruction that the student must not write code. The workspace contained the assignment but no implementation or configured Git remote.

The search focused on recent probabilistic image classification methods and primary research sources. Queries included "2024 2025 evidential deep learning image classification Dirichlet paper official" and "Fisher Information-based Evidential Deep Learning ICML 2023 paper". The agent considered I-EDL [4], HENN [5], and the variance-based evidential method [1].

The selected paper is **Evidential Uncertainty Quantification: A Variance-Based Perspective**, by Ruxiao Duan, Brian Caffo, Harrison X. Bai, Haris I. Sair, and Craig Jones, WACV 2024, pp. 2132-2141 [1]. Its preprint appeared in November 2023. The choice offers an explicit probability derivation, class-level outputs, and manageable local computation; it is not a claim that this is the newest method available in 2026.

The agent verified the conference metadata, read the primary paper through arXiv, and checked formulas against the authors' reference implementation [2]. The research trail and the exact implementation scope are recorded in docs/research_log.md.

Measured outcome. Over 3 seeds, the evidential classifier achieved 96.76% clean-test accuracy (softmax: 97.04%). Under severe pixel noise, evidential predictive NLL was 2.106, compared with 4.240 for softmax. These are this project's results, not reproduced paper benchmark numbers.

2. Algorithm and probability theory

The model uses a two-level probability model: the one-hot label y is categorical given a class-probability vector p, while p itself follows a Dirichlet distribution. A neural network predicts positive concentration parameters alpha. In this implementation alpha = softplus(logits) + 1, a smooth evidence transformation with a unit uniform prior.

```
y | p ~ Categorical(p) ; p | x ~ Dirichlet(alpha(x))
S = sum(alpha) ; m = E[p | x] = alpha / S
prediction = argmax(m)
```

The distinction matters: two images may have the same predictive mean m but different Dirichlet strength S. The larger strength represents greater concentration of the learned probability distribution. Network weights remain deterministic; this is a probability-based classifier rather than a Bayesian posterior over weights.

Training follows the paper's supervised objective (Eqs. 5-7): predictive negative log likelihood plus a KL penalty against a uniform Dirichlet. The target-class concentration is replaced by 1 only in the penalty, so regularization penalizes unsupported evidence for other classes.

```
L = log(S) - log(alpha_true) + lambda * KL
KL = KL(Dirichlet(masked_alpha) || Dirichlet(ones))
masked_alpha = y + (1-y) * alpha
```

The law of total covariance separates uncertainty under the fitted model [1]. Let C = diag(m) - m m^T. The paper's Eqs. 19-27 yield:

```
Cov(y | x) = C
Epistemic covariance = C / (S+1)
Aleatoric covariance = S*C / (S+1)
```

A covariance diagonal gives uncertainty for each class; its trace gives sample uncertainty. Correlation divides each covariance entry by the corresponding standard deviations (Eq. 28). Aleatoric uncertainty describes randomness conditional on the model; epistemic uncertainty describes dispersion of its predicted probability vector. These interpretations depend on the Dirichlet assumptions and training quality.

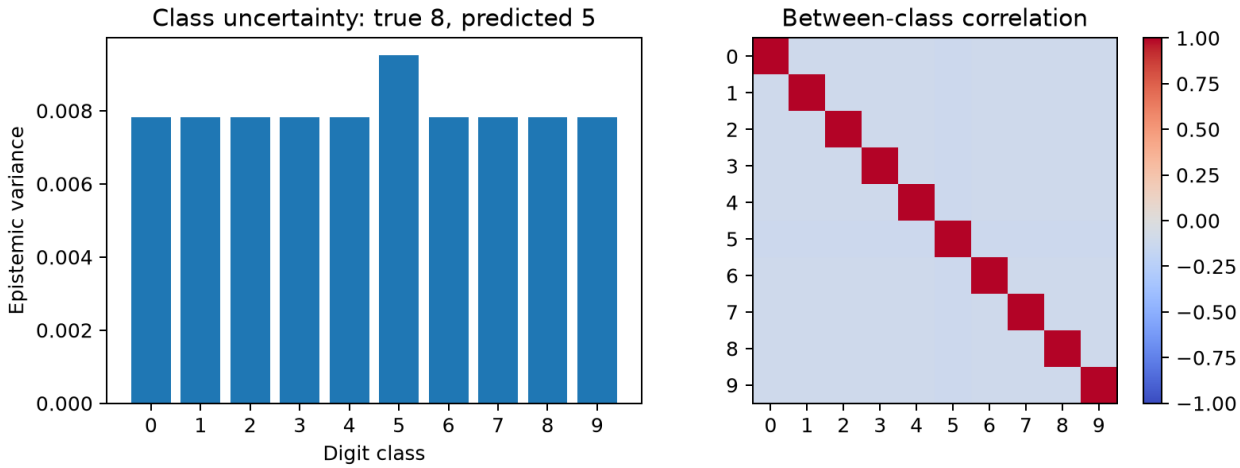


Figure 1. Seed 42: the clean test image with highest epistemic uncertainty is an 8 predicted as 5. Its broad class uncertainty reflects low evidence. Negative off-diagonal correlation comes from mutually exclusive labels; it is not an empirical confusion rate.

3. How AI implemented and verified it

The agent wrote an independent NumPy MLP with manual backpropagation and Adam. SciPy provides special functions for the KL loss and its analytic gradient. scikit-learn supplies the bundled digit data, stratified splits, and standard metrics; Matplotlib draws the figures. No code was copied from the reference repository.

Module	Purpose
src/evidential.py	Dirichlet loss, gradients, network, covariance, entropy, checkpoints
src/experiment.py	Paired experiments, validation selection, metrics, plots
src/predict.py	Prediction on a bundled digit or an 8x8 grayscale image
src/report.py	Generate this PDF directly from results/metrics.json

Scope. Implemented paper Eqs. 5-10 and 19-28. Adapted the backbone to a small MLP and the output to softplus + 1. The full active domain adaptation loop, unsupervised uncertainty guidance, certainty pseudo-labeling, and Office-Home/VisDA experiments were not implemented. The result is a classification/uncertainty adaptation, not a full reproduction.

Checks performed. Nine pytest cases passed. The analytic loss gradient agreed with central finite differences. Covariance components agreed with 150,000 sampled Dirichlet probability vectors. Other checks covered decomposition, covariance row sums and positive semidefiniteness, evidence scaling, invalid inputs, checkpoint round trips, training loss reduction, and disjoint split indices.

4. Experiment settings

Setting	Value
Dataset [3]	1,797 optical digit images; 8x8 pixels; 10 classes (not MNIST)
Preprocessing / splits	Pixels divided by 16; 1,077 train / 360 validation / 360 test
Backbone / optimizer	64-128-10 ReLU MLP; Adam; learning rate 0.003; batch 64
Training / selection	250 epochs; restore minimum clean validation predictive NLL
Evidential KL	Linear ramp to 0.01 over the first 10 epochs
Repeated runs	42, 43, 44
Test corruption	Gaussian noise SD 0.3 and 0.6 on normalized pixels; clip to [0,1]

Both methods use the same initialization, batches, splits, and test noise within a seed. Training uses clean data only; test images never guide checkpoint selection. Seeds vary both the split and training randomness. Mean +/- sample SD summarizes run variability; overlapping random splits do not constitute independent datasets or a significance test. All models, split indices, prediction arrays, and histories are saved.

4. Experiment results

Tables report mean +/- sample SD across the three runs. Accuracy and macro F1 are percentages. NLL is predictive negative log likelihood; lower is better. The Brier score sums squared probability errors over all 10 classes. ECE uses 15 equal-width confidence bins; it is a finite-sample calibration estimate.

Condition / method	Accuracy (%)	Macro F1 (%)	NLL
Clean / softmax	97.04 +/- 0.16	97.04 +/- 0.14	0.092 +/- 0.012
Clean / evidential	96.76 +/- 0.42	96.75 +/- 0.42	0.162 +/- 0.010
Noise 0.3 / softmax	80.56 +/- 3.74	80.18 +/- 4.23	0.800 +/- 0.295
Noise 0.3 / evidential	80.28 +/- 1.55	79.79 +/- 1.88	0.740 +/- 0.043
Noise 0.6 / softmax	43.15 +/- 2.94	42.36 +/- 3.37	4.240 +/- 1.340
Noise 0.6 / evidential	44.26 +/- 1.05	42.66 +/- 1.28	2.106 +/- 0.058

Condition / method	Brier score	ECE
Clean / softmax	0.044 +/- 0.002	0.017 +/- 0.005
Clean / evidential	0.055 +/- 0.005	0.073 +/- 0.009
Noise 0.3 / softmax	0.305 +/- 0.071	0.117 +/- 0.042
Noise 0.3 / evidential	0.305 +/- 0.018	0.065 +/- 0.012
Noise 0.6 / softmax	0.961 +/- 0.087	0.434 +/- 0.052
Noise 0.6 / evidential	0.832 +/- 0.015	0.279 +/- 0.020

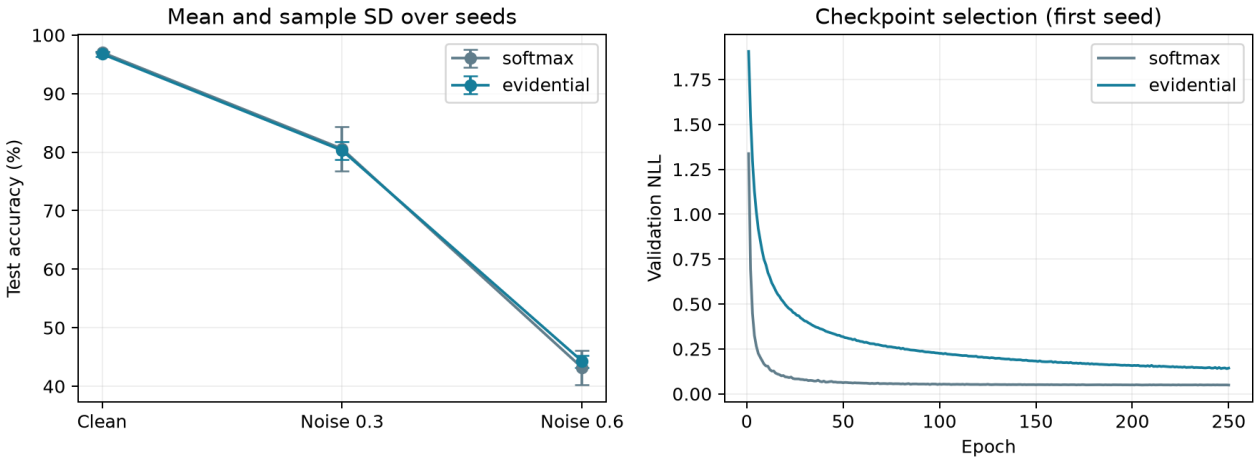


Figure 2. Accuracy declines under corruption for both methods. The right plot shows the first seed's validation NLL over training; each method restores its own best epoch.

Softmax performs better on clean NLL, Brier score, and ECE. The evidential model has lower mean noisy NLL and ECE here, but moderate-noise accuracy is slightly lower. The severe-noise accuracy advantage is small compared with run variability. These observations support a limited probabilistic robustness benefit, not universal superiority.

4. Uncertainty analysis and limitations

Variance-based and entropy-based uncertainty use the same trained evidential network; changing the uncertainty formula does not change digit predictions. Their error-detection AUROC treats a mistake as the positive class. A larger score should rank incorrect images above correct images; 0.5 represents chance ranking. This is not an out-of-distribution detection experiment.

Condition	Variance EU AUROC	Entropy EU AUROC	Mean AU / EU
Clean	0.970 +/- 0.017	0.932 +/- 0.041	0.1593 / 0.0020
Noise 0.3	0.860 +/- 0.010	0.816 +/- 0.035	0.3080 / 0.0076
Noise 0.6	0.673 +/- 0.049	0.658 +/- 0.047	0.3908 / 0.0091

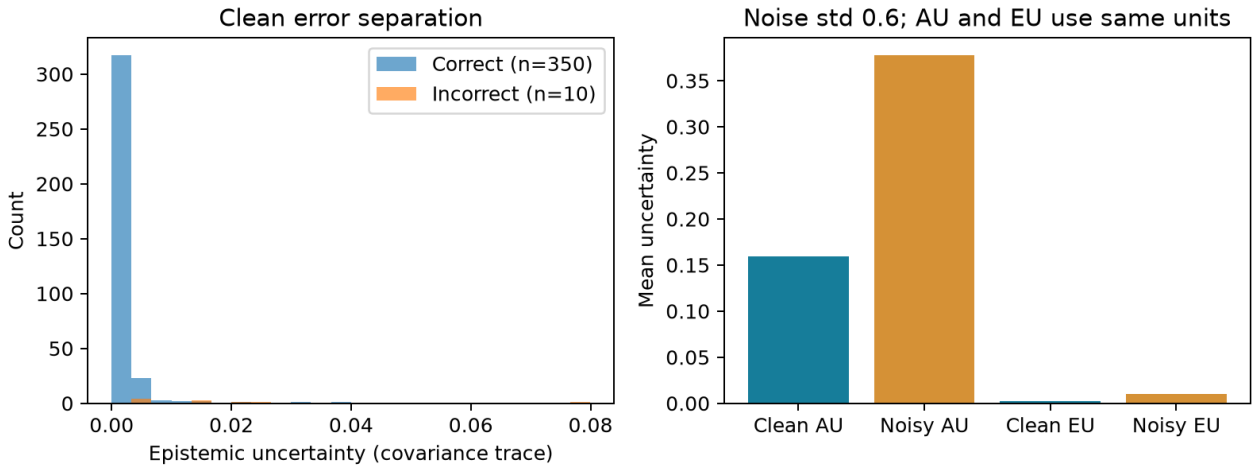


Figure 3. Seed 42 error-score distributions and clean/severely noisy uncertainty means. Table entries above aggregate all three seeds; the plots use one seed. AU and EU use covariance-trace units.

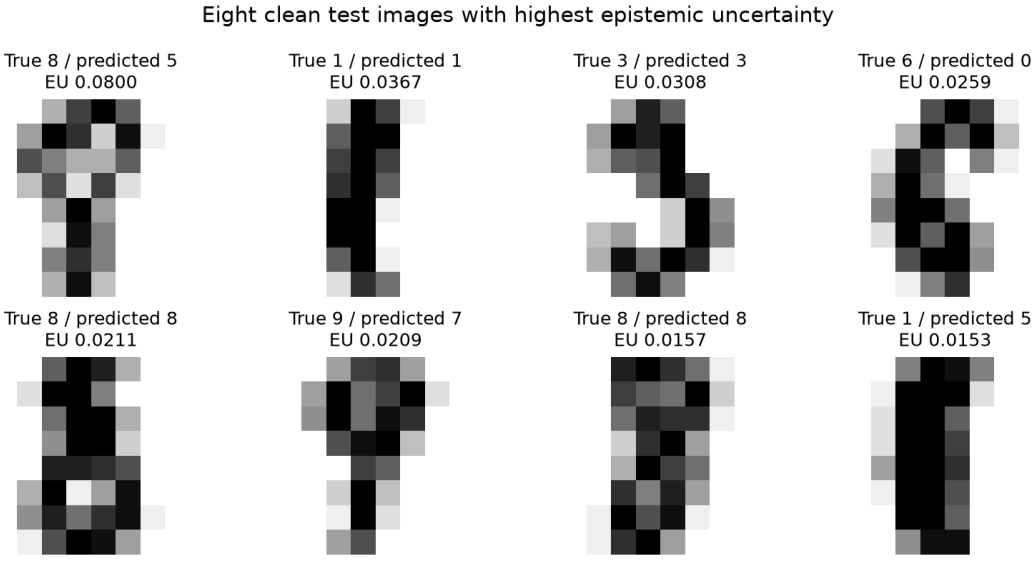


Figure 4. Seed 42's eight most epistemically uncertain clean test images, including correct and incorrect predictions. High uncertainty does not imply that a prediction must be wrong.

Both mean AU and EU increase under corruption, and variance EU ranks errors better than entropy EU in these runs. Severe-noise error ranking remains limited (about 0.67 AUROC), and evidential ECE remains high. The small dataset, fixed hyperparameters, random image splits, and synthetic noise limit generalization. Digit pixels do not approximate the paper's large cross-domain image benchmark. No automatic rejection threshold was tuned.

5. What I learnt from this AI assignment

Personal learning reflection:

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6. Webpage link of my source code

Source-code webpage: _____

Work remains local at the student's request. This field is intentionally blank. A webpage for this project's source code is required before submission; the authors' reference repository does not replace it.

References and reproducibility

[1] Duan, R., Caffo, B., Bai, H. X., Sair, H. I., and Jones, C. (2024). Evidential Uncertainty Quantification: A Variance-Based Perspective. WACV, 2132-2141. Primary paper: https://arxiv.org/abs/2311.11367

[2] Duan et al. Authors' reference implementation: https://github.com/KerryDRX/EvidentialADA

[3] scikit-learn. load_digits: bundled optical recognition of handwritten digits: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_digits.html

[4] Deng et al. (2023). Uncertainty Estimation by Fisher Information-based Evidential Deep Learning. ICML: https://proceedings.mlr.press/v202/deng23b.html

[5] Li, C., et al. (2024). Hyper Evidential Deep Learning to Quantify Composite Classification Uncertainty. ICLR 2024. Considered as an alternative during algorithm selection.

Reproduce: .venv/bin/python -m pytest -q ; .venv/bin/python -m src.experiment ; .venv/bin/python -m src.report. Exact dependencies are recorded in requirements-lock.txt. Detailed AI steps and publication URLs are in docs/research_log.md. Raw measurements are in results/metrics.json and results/metrics.csv.

Report status: local draft. Student details, personal reflection, and publication link require completion. No public publishing or Moodle submission has been performed.